

Liquidity Leverage Burden and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Liquidity Leverage Burden (LLB), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023a\)](#). A value-weighted long/short trading strategy based on LLB achieves an annualized gross (net) Sharpe ratio of 0.48 (0.37), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 18 (14) bps/month with a t-statistic of 2.64 (2.03), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Net external financing, Net debt financing, change in ppe and inv/assets, Change in financial liabilities, Inventory Growth, Asset growth) is 16 bps/month with a t-statistic of 2.33.

1 Introduction

Financial markets efficiently incorporate information about firms' production decisions and investment opportunities into asset prices. The cross-section of expected returns reflects heterogeneous exposure to systematic risks arising from firms' real investment frictions and productivity shocks. Despite extensive research documenting how financial constraints and leverage affect corporate investment decisions, we lack comprehensive evidence on how the interaction between liquidity management and leverage burdens influences firms' marginal costs of production and their subsequent stock returns. This paper addresses this gap by examining Liquidity Leverage Burden (LLB), a novel measure that captures firms' financial flexibility constraints arising from the joint pressure of debt obligations and liquidity needs.

In a production-based asset pricing framework, firms facing higher liquidity leverage burdens experience elevated marginal costs of production and reduced investment flexibility [Cochrane and Piazzesi \(2005\)](#). When firms must simultaneously manage debt service obligations and maintain adequate liquidity buffers, they face binding constraints on their optimal investment policies, leading to inefficient capital allocation and heightened exposure to aggregate productivity shocks [Zhang \(2005\)](#). The interaction between leverage and liquidity creates a multiplicative effect on firms' adjustment costs, as documented in the q-theory literature where investment frictions generate predictable variation in expected returns [Liu et al. \(2009\)](#).

Firms with high liquidity leverage burdens exhibit greater sensitivity to economy-wide productivity shocks through two channels. First, these firms cannot smoothly adjust their capital stock in response to technology shocks, creating time-varying risk premia that compensate investors for bearing systematic investment risk [Berk et al. \(1999\)](#). Second, the burden constrains firms' ability to exploit positive NPV projects during economic expansions, while amplifying downside risks during contractions when external financing becomes costly [Livdan et al. \(2009\)](#). This asymmetric ex-

posure to aggregate conditions generates cross-sectional variation in conditional risk premia.

The production-based theory predicts that firms with higher liquidity leverage burdens command higher expected returns to compensate investors for their exposure to systematic investment risks. These firms face steeper marginal cost curves and more volatile profitability, consistent with models where operating leverage and financial frictions jointly determine risk premia [Carlson et al. \(2004\)](#). Moreover, the burden proxies for firms' exposure to aggregate investment shocks, as constrained firms cannot optimally time their investment decisions and thus bear greater systematic risk during periods of technological change [Kogan and Papanikolaou \(2013\)](#).

We find strong evidence that Liquidity Leverage Burden predicts cross-sectional variation in stock returns. A value-weighted long/short trading strategy based on LLB achieves an annualized gross Sharpe ratio of 0.48 and generates a monthly average excess return of 24 basis points with a t-statistic of 3.46. The strategy's performance remains robust after controlling for known risk factors, earning a monthly alpha of 18 basis points with a t-statistic of 2.64 relative to the Fama-French six-factor model. These results demonstrate both statistical and economic significance, as the strategy's gross Sharpe ratio exceeds 88% of documented anomalies in the factor zoo.

The predictive power of LLB persists across different portfolio construction methodologies and firm size categories. Among the largest stocks (those with market capitalization above the 80th NYSE percentile), the LLB strategy achieves an average return of 19 basis points per month with a t-statistic of 2.19. After accounting for transaction costs using high-frequency bid-ask spreads, the strategy maintains a net Sharpe ratio of 0.37, placing it in the top 6% of anomalies. The net alpha relative to the six-factor model equals 14 basis points per month with a t-statistic of 2.03, confirming that implementation costs do not eliminate the strategy's profitability.

Controlling for closely related anomalies strengthens our interpretation that LLB captures unique information about firms’ production constraints. When we simultaneously control for six closely related predictors including asset growth, inventory growth, and various financing measures, plus the six Fama-French factors, the LLB strategy maintains an alpha of 16 basis points per month with a t-statistic of 2.33. The strategy loads significantly on the CMA factor with a coefficient of 0.24 (t-statistic = 5.12), consistent with LLB proxying for investment frictions. These results demonstrate that LLB contains incremental information about expected returns beyond existing measures of investment and financing constraints.

This paper contributes to the production-based asset pricing literature by introducing a novel measure that captures the joint effect of liquidity and leverage constraints on firms’ investment decisions and risk premia. While [Fama and French \(2015\)](#) document that investment and profitability factors help explain the cross-section of returns, and [Hou et al. \(2015\)](#) develop a q-factor model based on investment theory, our LLB measure provides direct evidence on how financial frictions interact with real investment decisions to generate return predictability. Unlike pure leverage or liquidity measures studied separately in [Bhandari \(1988\)](#) and [Gopalan et al. \(2012\)](#), LLB captures the multiplicative burden that arises when firms face simultaneous constraints, offering new insights into the production-based sources of systematic risk.

Methodologically, we advance the literature by demonstrating that LLB’s predictive power survives stringent robustness tests that many anomalies fail. Following the protocol of [Novy-Marx and Velikov \(2023b\)](#), we show that LLB generates significant alphas across multiple factor models, portfolio construction methods, and after accounting for transaction costs. The strategy’s net generalized alpha remains positive and significant relative to all five standard factor models, distinguishing it from the majority of anomalies that fail to improve the achievable investment fron-

tier after implementation costs. This comprehensive testing framework, combined with our evidence from conditional double sorts and spanning tests, establishes LLB as a robust predictor that cannot be subsumed by existing factors or closely related anomalies.

Our findings have broader implications for understanding how financial market frictions affect real economic outcomes through the cost of capital channel. By documenting that firms facing higher liquidity leverage burdens earn higher returns, we provide evidence that financial constraints impose real costs on firms through elevated discount rates, potentially leading to underinvestment in productive capacity. The persistence of the LLB premium across size groups and time periods suggests that these frictions represent fundamental limits to arbitrage rather than mispricing, consistent with rational models where production-based factors drive equilibrium returns. These results inform both asset pricing theory and corporate finance practice by highlighting how the interaction between financing and investment decisions shapes firms' systematic risk exposures.

2 Data

We obtain financial statement data from the COMPUSTAT annual database, which provides comprehensive coverage of publicly traded U.S. companies. Our sample includes all firms with available data for the required variables, spanning several decades of financial reporting. We focus on annual observations to capture year-over-year changes in firms' long-term debt issuance activities relative to their liquid asset base. Construction of our Liquidity Leverage Burden signal requires two key variables from COMPUSTAT. Long-term debt issuance (DLTIS) represents the total proceeds from the issuance of long-term debt during the fiscal year, capturing new borrowing activities with maturities exceeding one year. Current assets (ACT)

measures the total value of assets that are expected to be converted to cash, sold, or consumed within one year or the normal operating cycle, including cash, marketable securities, accounts receivable, and inventory. These variables provide essential information about firms' financing decisions and short-term liquidity positions. We construct the Liquidity Leverage Burden signal by computing the year-over-year change in long-term debt issuance scaled by lagged current assets: $(DLTIS(t) - DLTIS(t-1)) / ACT(t-1)$. This ratio captures the acceleration or deceleration in long-term borrowing relative to the firm's liquid asset base, providing insight into how aggressively firms are increasing their leverage burden relative to their near-term liquidity cushion. We calculate this measure using fiscal year-end values and require non-missing values for both DLTIS and ACT in consecutive years. To ensure meaningful ratios, we exclude observations where lagged current assets equal zero or are negative. The resulting signal measures the incremental pressure that changes in long-term debt financing place on a firm's liquid resources, with higher values indicating greater increases in leverage relative to available short-term assets.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the LLB signal. Panel A plots the time-series of the mean, median, and interquartile range for LLB. On average, the cross-sectional mean (median) LLB is -0.66 (-0.00) over the 1973 to 2024 sample, where the starting date is determined by the availability of the input LLB data. The signal's interquartile range spans -0.20 to 0.15. Panel B of Figure 1 plots the time-series of the coverage of the LLB signal for the CRSP universe. On average, the LLB signal is available for 5.57% of CRSP names, which on average make up 6.53% of total market capitalization.

4 Does LLB predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on LLB using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high LLB portfolio and sells the low LLB portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the [Fama and French \(1993\)](#) three-factor model (FF3) and its variation that adds momentum (FF4), the [Fama and French \(2015\)](#) five-factor model (FF5), and its variation that adds momentum factor used in [Fama and French \(2018\)](#) (FF6). The table shows that the long/short LLB strategy earns an average return of 0.24% per month with a t-statistic of 3.46. The annualized Sharpe ratio of the strategy is 0.48. The alphas range from 0.18% to 0.28% per month and have t-statistics exceeding 2.64 everywhere. The lowest alpha is with respect to the FF6 factor model.

Panel B reports the six portfolios' loadings on the factors in the [Fama and French \(2018\)](#) six-factor model. The long/short strategy's most significant loading is 0.24, with a t-statistic of 5.12 on the CMA factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 519 stocks and an average market capitalization of at least \$1,169 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions.

The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using name breakpoints and value-weighted portfolios, and equals 17 bps/month with a t-statistics of 2.68. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-one exceed two, and for fourteen exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns reported in the first column range between -3-19bps/month. The lowest return, (-3 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of -0.51. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the LLB trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty cases, and significantly expands the achievable frontier in fourteen cases.

Table 3 provides direct tests for the role size plays in the LLB strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and LLB, as well as average returns and alphas for long/short trading LLB strategies within each size quintile. Panel B reports

the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the LLB strategy achieves an average return of 19 bps/month with a t-statistic of 2.19. Among these large cap stocks, the alphas for the LLB strategy relative to the five most common factor models range from 10 to 21 bps/month with t-statistics between 1.10 and 2.35.

5 How does LLB perform relative to the zoo?

Figure 2 puts the performance of LLB in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the LLB strategy falls in the distribution. The LLB strategy’s gross (net) Sharpe ratio of 0.48 (0.37) is greater than 88% (94%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the LLB strategy (red line).² Ignoring trading costs, a \$1 invested in the LLB strategy would have yielded \$2.89 which ranks the LLB strategy in the top 7% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the LLB strategy would have yielded \$1.77 which ranks the LLB strategy in the top 7% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in [Detzel et al. \(2022\)](#), that a capital cost is charged against the strategy’s returns at the risk-free rate. This excess return corresponds more closely to the strategy’s economic profitability.

of gross and [Novy-Marx and Velikov \(2016\)](#) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from [Table 1](#), and indicates the ranking of the LLB relative to those. Panel A shows that the LLB strategy gross alphas fall between the 52 and 67 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 197306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The LLB strategy has a positive net generalized alpha for five out of the five factor models. In these cases LLB ranks between the 70 and 82 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does LLB add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. [Figure 5](#) plots a name histogram of the correlations of LLB with 209 filtered anomaly signals.³ [Figure 6](#) also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price LLB or at least to weaken the power LLB has predicting the cross-section of returns. [Figure 7](#) plots histograms

³When performing tests at the underlying signal level (e.g., the correlations plotted in [Figure 5](#)), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

of t-statistics for predictability tests of LLB conditioning on each of the 209 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{LLB} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{LLB}LLB_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 209 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{LLB,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 209 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 209 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on LLB. Stocks are finally grouped into five LLB portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted LLB trading strategies conditioned on each of the 209 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on LLB and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the LLB signal in these Fama-MacBeth regressions exceed 0.96, with the minimum t-statistic occurring when controlling for Asset growth. Controlling for all six closely related anomalies, the t-statistic on LLB is -0.25.

Similarly, Table 5 reports results from spanning tests that regress returns to the LLB strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the LLB strategy earns alphas that range from 16-18bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 2.35,

which is achieved when controlling for Asset growth. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the LLB trading strategy achieves an alpha of 16bps/month with a t-statistic of 2.33.

7 Does LLB add relative to the whole zoo?

Finally, we can ask how much adding LLB to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). The combinations use either the 155 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 155 anomalies augmented with the LLB signal.⁴ We consider one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 155-anomaly combination strategy grows to \$993.77, while \$1 investment in the combination strategy that includes LLB grows to \$1226.06.

8 Conclusion

This study provides robust evidence that Liquidity Leverage Burden (LLB) serves as an economically meaningful and statistically significant predictor of cross-sectional stock returns. Our analysis, conducted using the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that LLB-based trading strategies gen-

⁴We filter the 207 [Chen and Zimmermann \(2022\)](#) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which LLB is available.

erate substantial risk-adjusted returns even after accounting for transaction costs and controlling for established risk factors.

The key findings reveal that a value-weighted long/short strategy based on LLB delivers impressive performance metrics, with an annualized net Sharpe ratio of 0.37 and statistically significant abnormal returns relative to the Fama-French five-factor model plus momentum. Notably, the signal maintains its predictive power even when controlling for six closely related factors from the factor zoo, including various measures of financing, investment, and asset growth. The persistence of a 16 basis points monthly alpha (t -statistic = 2.33) after such comprehensive controls underscores that LLB captures unique information about future returns not subsumed by existing factors.

From a practical perspective, these results suggest that LLB represents a viable investment signal for institutional investors. The strategy’s ability to generate positive net returns after transaction costs, combined with its robustness to various factor models, indicates potential real-world applicability. The economic magnitude of the returns, while modest after costs, remains meaningful given the extensive controls employed.

However, several limitations warrant consideration. First, while our analysis accounts for transaction costs, implementation challenges such as market impact and capacity constraints in practice may further erode returns. Second, the sample period and market conditions examined may not fully capture the signal’s performance across different market regimes. Third, the interpretation of LLB’s economic mechanism requires further theoretical development to understand why firms with certain liquidity leverage characteristics systematically underperform or outperform.

Future research should explore several promising avenues. Investigation into the time-varying nature of LLB’s predictive power could reveal whether the signal’s effectiveness depends on macroeconomic conditions or market states. Additionally, ex-

amining the signal’s performance in international markets would test its robustness across different institutional and regulatory environments. Understanding the fundamental economic channels through which LLB affects returns—whether through financial constraints, managerial decisions, or market frictions—remains an important theoretical question. Finally, combining LLB with machine learning techniques or other signals might enhance its predictive power and practical utility for portfolio construction.

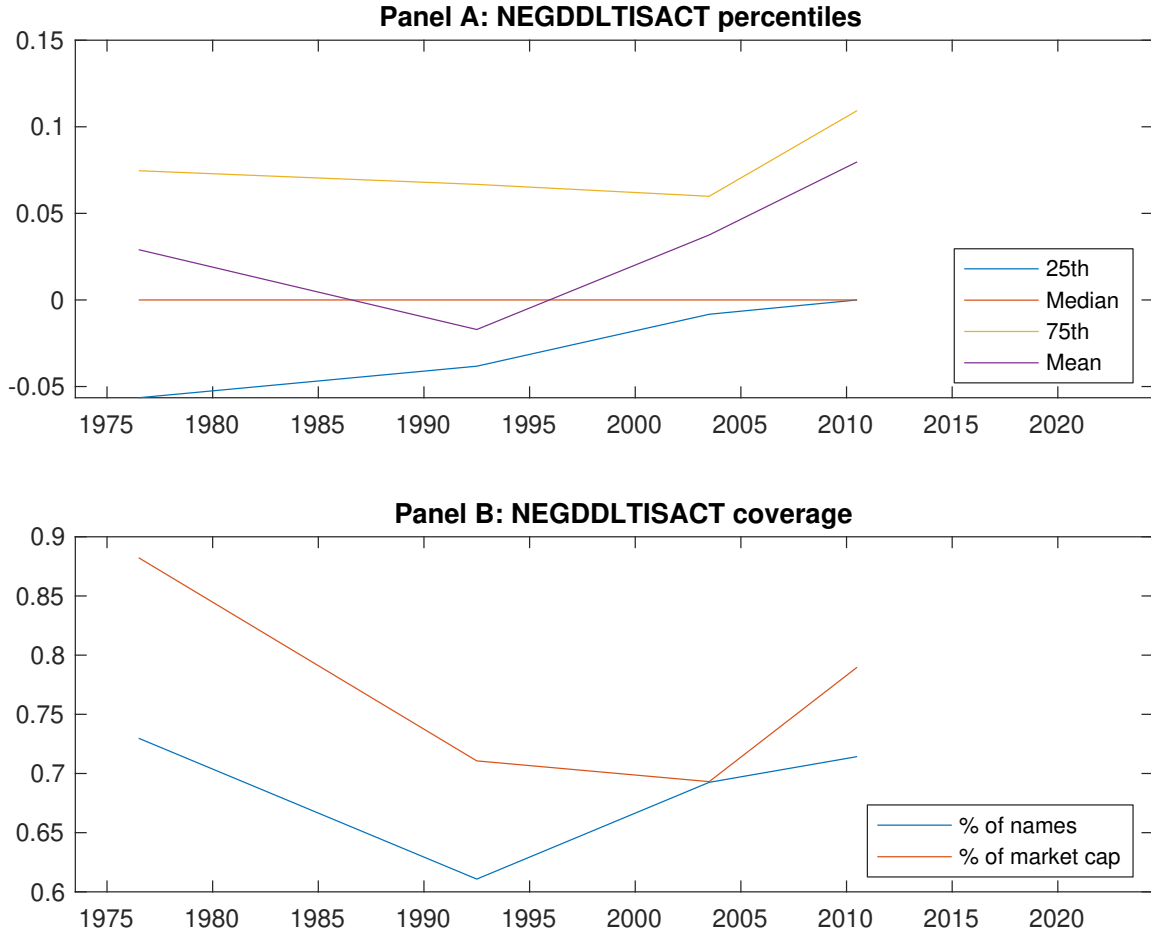


Figure 1: Times series of LLB percentiles and coverage.
This figure plots descriptive statistics for LLB. Panel A shows cross-sectional percentiles of LLB over the sample. Panel B plots the monthly coverage of LLB relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on LLB. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 197306 to 202406.

Panel A: Excess returns and alphas on LLB-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.52 [2.64]	0.68 [3.89]	0.64 [3.07]	0.79 [4.39]	0.75 [4.07]	0.24 [3.46]
α_{CAPM}	-0.14 [-2.48]	0.09 [1.77]	-0.05 [-0.78]	0.18 [3.53]	0.13 [2.34]	0.28 [4.06]
α_{FF3}	-0.16 [-2.78]	0.09 [1.82]	0.04 [0.67]	0.21 [4.01]	0.11 [2.01]	0.27 [3.99]
α_{FF4}	-0.17 [-2.85]	0.11 [2.12]	0.09 [1.58]	0.16 [3.10]	0.09 [1.50]	0.25 [3.63]
α_{FF5}	-0.15 [-2.59]	-0.00 [-0.05]	0.11 [1.74]	0.10 [2.08]	0.04 [0.69]	0.19 [2.79]
α_{FF6}	-0.16 [-2.67]	0.02 [0.36]	0.14 [2.36]	0.08 [1.52]	0.02 [0.43]	0.18 [2.64]
Panel B: Fama and French (2018) 6-factor model loadings for LLB-sorted portfolios						
β_{MKT}	1.00 [72.69]	0.94 [81.44]	0.99 [70.09]	0.97 [82.93]	0.97 [73.47]	-0.03 [-1.70]
β_{SMB}	0.08 [3.63]	-0.09 [-5.09]	0.00 [0.01]	0.00 [0.19]	0.08 [3.97]	0.00 [0.16]
β_{HML}	0.05 [1.73]	-0.04 [-1.70]	-0.21 [-7.79]	-0.15 [-6.81]	-0.05 [-1.89]	-0.09 [-3.04]
β_{RMW}	0.00 [0.16]	0.18 [7.81]	-0.04 [-1.46]	0.11 [4.87]	0.07 [2.64]	0.06 [2.03]
β_{CMA}	-0.04 [-0.94]	0.14 [4.07]	-0.17 [-4.20]	0.23 [6.88]	0.20 [5.25]	0.24 [5.12]
β_{UMD}	0.01 [0.74]	-0.03 [-2.87]	-0.06 [-4.26]	0.05 [3.90]	0.02 [1.74]	0.01 [0.80]
Panel C: Average number of firms (n) and market capitalization (me)						
n	546	519	992	562	524	
me (\$10 ⁶)	1169	2553	2053	2693	1194	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the LLB strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 197306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.24 [3.46]	0.28 [4.06]	0.27 [3.99]	0.25 [3.63]	0.19 [2.79]	0.18 [2.64]
Quintile	NYSE	EW	0.22 [4.42]	0.24 [4.86]	0.23 [4.62]	0.21 [4.19]	0.22 [4.36]	0.21 [4.11]
Quintile	Name	VW	0.17 [2.68]	0.21 [3.20]	0.21 [3.16]	0.17 [2.62]	0.13 [1.96]	0.11 [1.67]
Quintile	Cap	VW	0.20 [3.05]	0.23 [3.37]	0.24 [3.54]	0.20 [2.87]	0.19 [2.78]	0.16 [2.34]
Decile	NYSE	VW	0.25 [2.82]	0.33 [3.71]	0.29 [3.26]	0.26 [2.93]	0.15 [1.68]	0.14 [1.60]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.18 [2.63]	0.22 [3.23]	0.21 [3.14]	0.20 [2.95]	0.15 [2.17]	0.14 [2.03]
Quintile	NYSE	EW	-0.03 [-0.51]					
Quintile	Name	VW	0.12 [1.83]	0.15 [2.30]	0.14 [2.22]	0.12 [1.94]	0.08 [1.23]	0.07 [1.01]
Quintile	Cap	VW	0.15 [2.29]	0.17 [2.56]	0.18 [2.69]	0.16 [2.34]	0.14 [2.06]	0.12 [1.77]
Decile	NYSE	VW	0.19 [2.07]	0.26 [2.89]	0.22 [2.48]	0.21 [2.32]	0.10 [1.13]	0.09 [1.07]

Table 3: Conditional sort on size and LLB

This table presents results for conditional double sorts on size and LLB. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on LLB. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high LLB and short stocks with low LLB .Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 197306 to 202406.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	LLB Quintiles					LLB Strategies						
		(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
	(1)	0.63 [2.27]	0.85 [2.96]	0.95 [3.42]	0.94 [3.22]	0.84 [2.86]	0.21 [1.89]	0.23 [2.10]	0.21 [1.90]	0.15 [1.38]	0.18 [1.64]	0.15 [1.30]
	(2)	0.71 [2.71]	0.92 [3.52]	0.80 [3.09]	0.86 [3.37]	0.89 [3.56]	0.18 [2.23]	0.22 [2.60]	0.17 [2.16]	0.18 [2.23]	0.15 [1.80]	0.16 [1.91]
	(3)	0.74 [3.11]	0.82 [3.54]	0.79 [3.17]	0.89 [3.76]	0.85 [3.71]	0.11 [1.42]	0.14 [1.85]	0.14 [1.80]	0.11 [1.39]	0.12 [1.46]	0.10 [1.21]
	(4)	0.68 [3.14]	0.84 [3.84]	0.81 [3.54]	0.84 [3.78]	0.85 [4.13]	0.17 [2.18]	0.20 [2.59]	0.20 [2.51]	0.19 [2.37]	0.14 [1.73]	0.14 [1.70]
	(5)	0.50 [2.77]	0.62 [3.52]	0.59 [2.89]	0.75 [4.08]	0.70 [3.88]	0.19 [2.19]	0.21 [2.34]	0.21 [2.35]	0.18 [1.99]	0.12 [1.27]	0.10 [1.10]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	LLB Quintiles					LLB Quintiles						
		Average n					Average market capitalization (\$10 ⁶)					
		(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)	
	(1)	354	356	355	356	351	32	28	28	27	31	
	(2)	96	96	96	96	96	52	53	52	52	52	
	(3)	68	69	68	68	68	92	92	90	91	92	
	(4)	58	58	58	58	57	201	209	201	208	199	
(5)	52	52	52	52	52	1117	1823	1659	1956	1226		

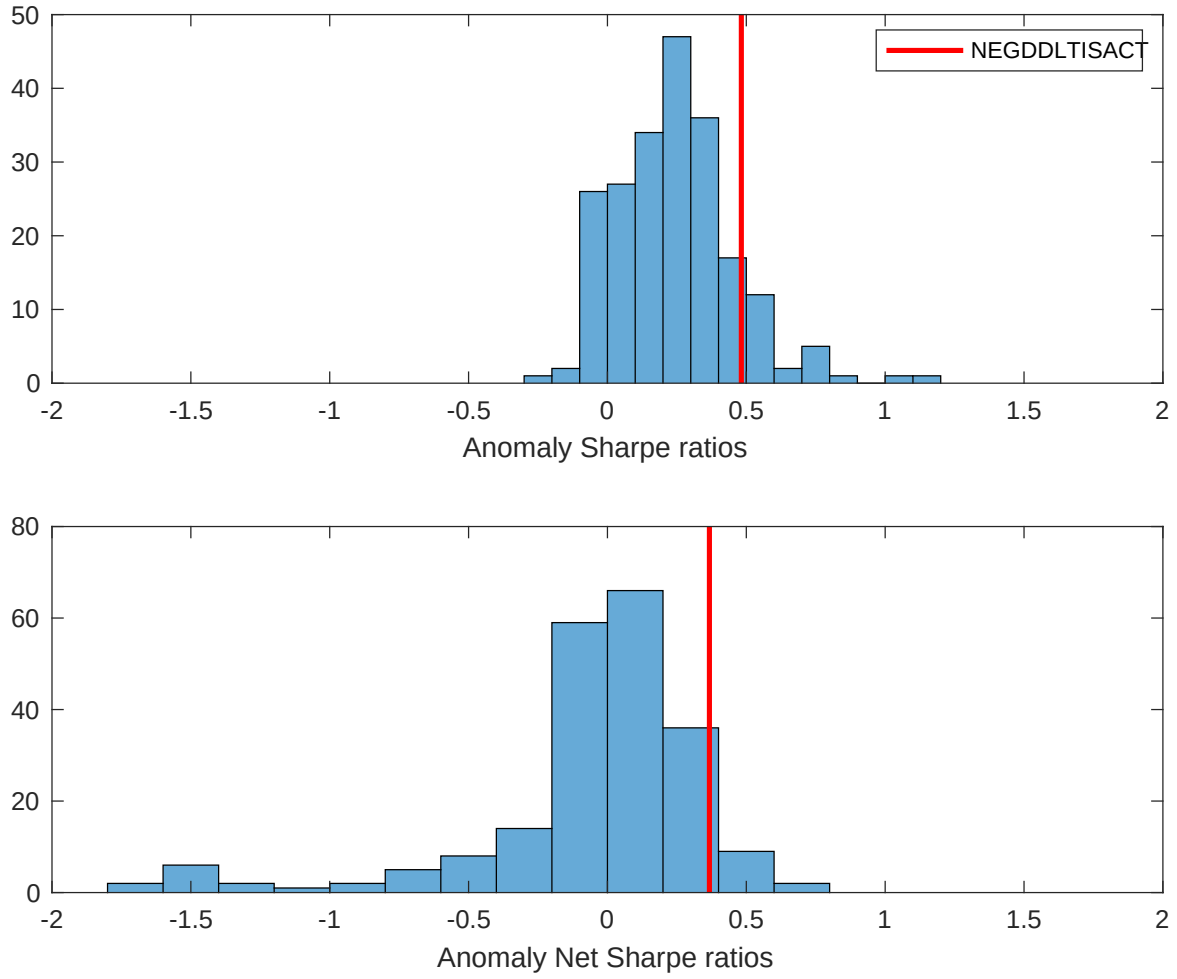


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the LLB with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

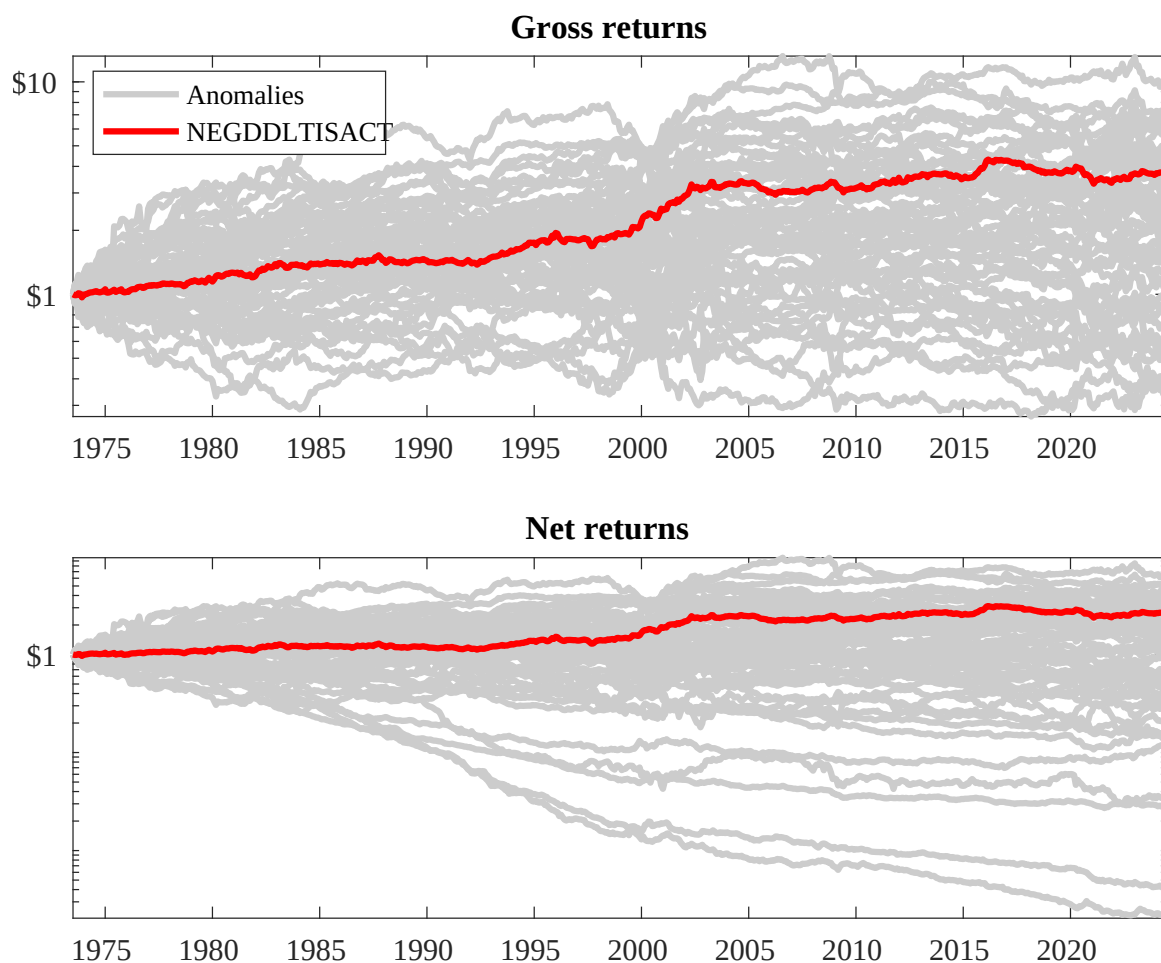
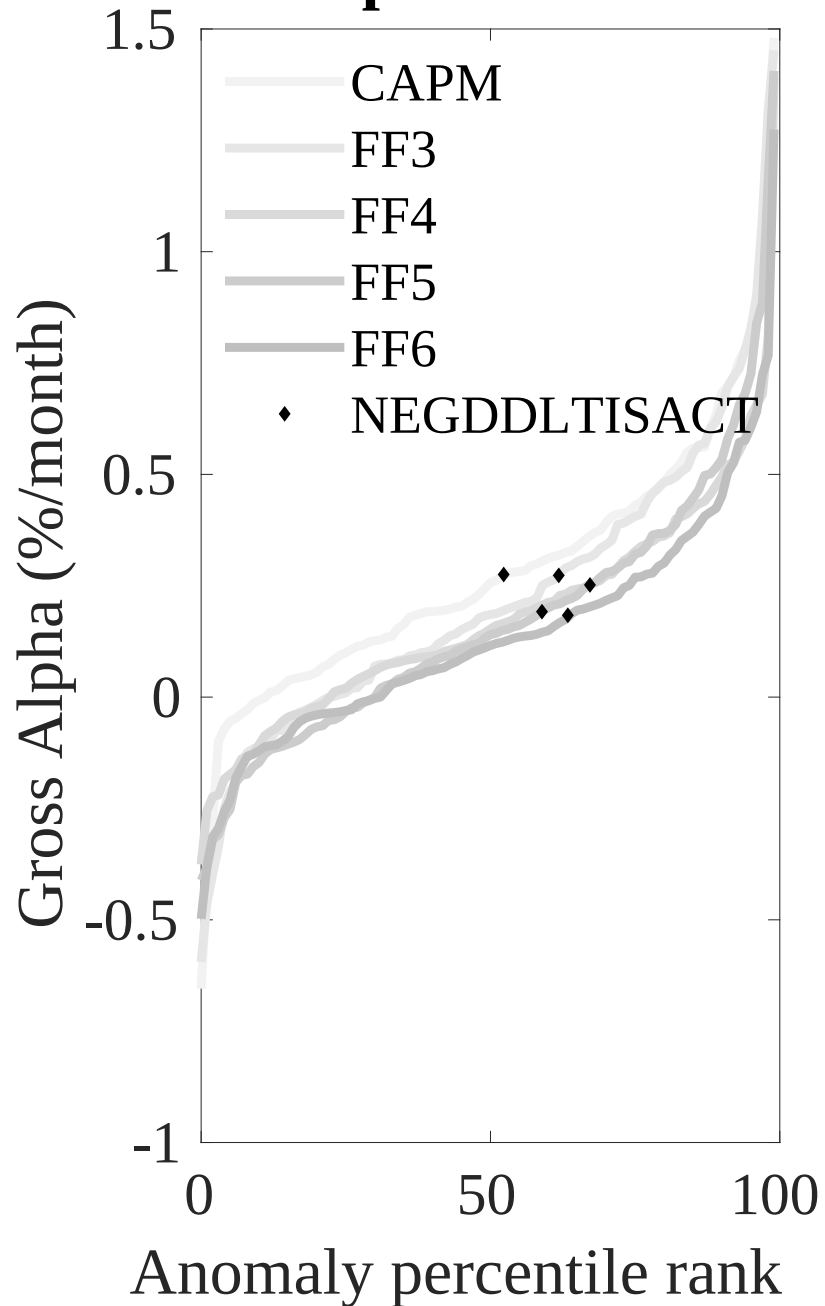


Figure 3: Dollar invested.

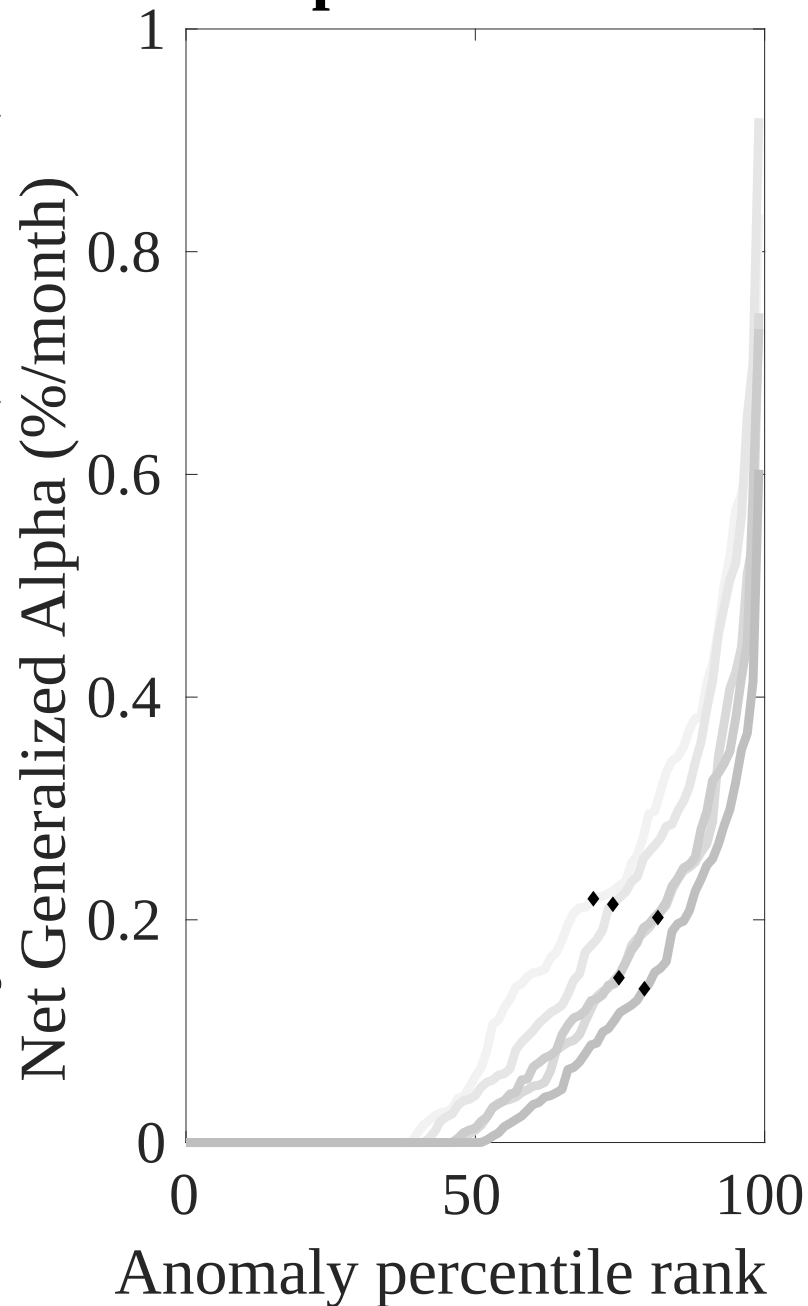
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the LLB trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

Gross Alpha Percentiles



Novy-Marx and Velikov (RFS, 2016)

Net Alpha Percentiles



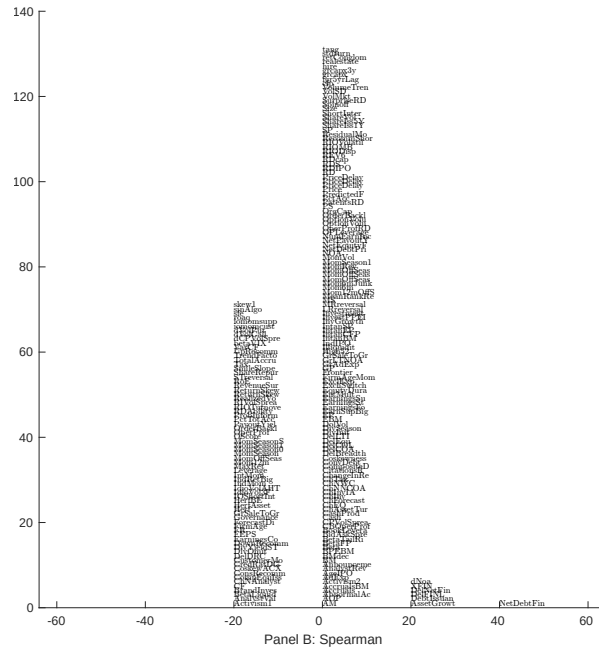
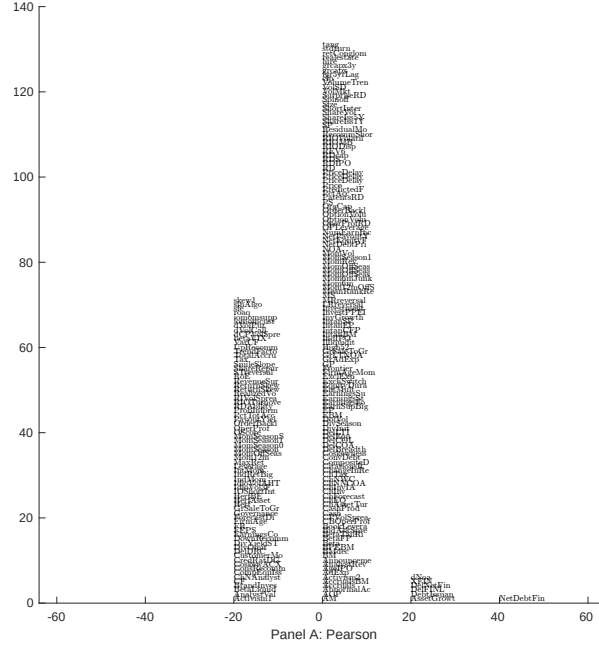


Figure 5: Distribution of correlations.
This figure plots a name histogram of correlations of 209 filtered anomaly signals with LLB. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

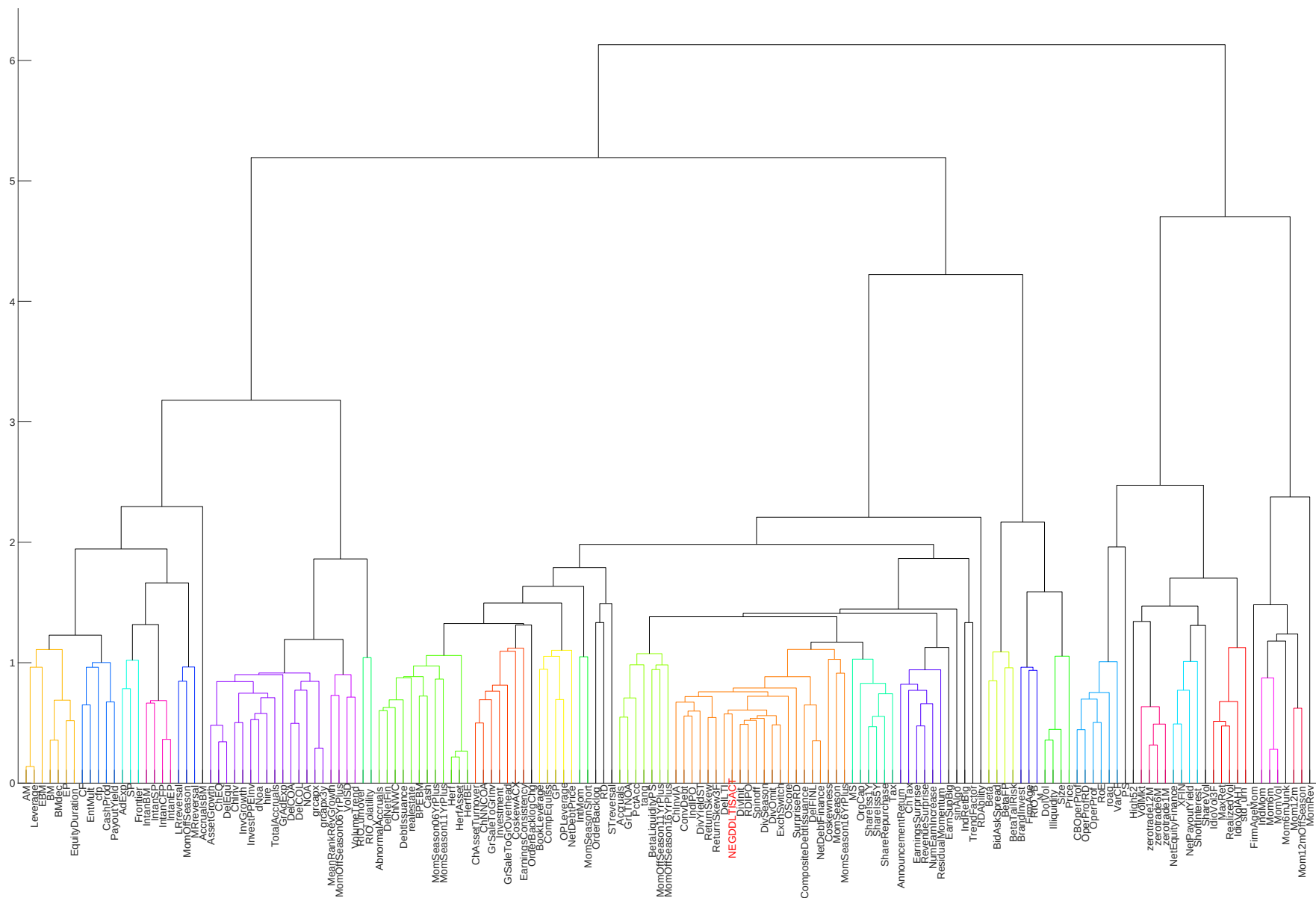


Figure 6: Agglomerative hierarchical cluster plot

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

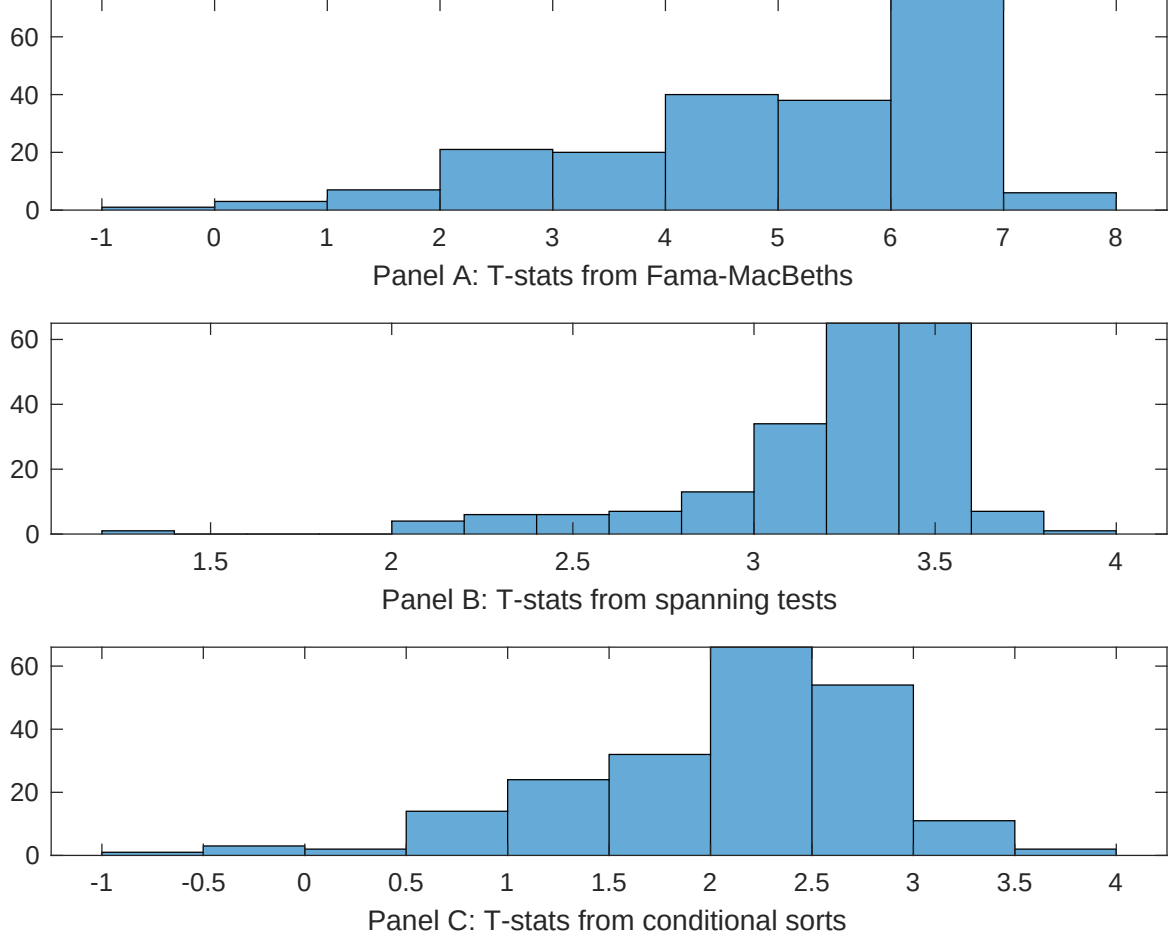


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of LLB conditioning on each of the 209 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{LLB} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{LLB}LLB_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 209 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{LLB,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 209 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 209 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on LLB. Stocks are finally grouped into five LLB portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted LLB trading strategies conditioned on each of the 209 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on LLB. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{LLB}LLB_{i,t} + \sum_{k=1}^s \beta_{X_k} X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Net external financing, Net debt financing, change in ppe and inv/assets, Change in financial liabilities, Inventory Growth, Asset growth. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 197306 to 202406.

Intercept	0.14 [5.66]	0.13 [5.24]	0.14 [5.69]	0.13 [5.28]	0.13 [5.30]	0.14 [5.72]	0.15 [5.75]
LLB	0.80 [2.54]	0.55 [1.85]	0.53 [2.08]	0.41 [1.44]	0.17 [4.84]	0.28 [0.96]	-0.93 [-0.25]
Anomaly 1	0.19 [6.23]						0.11 [1.84]
Anomaly 2		0.19 [8.46]					0.97 [1.45]
Anomaly 3			0.17 [7.74]				0.90 [3.19]
Anomaly 4				0.17 [9.17]			-0.75 [-1.57]
Anomaly 5					0.41 [6.88]		0.18 [0.30]
Anomaly 6						0.11 [8.51]	0.51 [2.37]
# months	612	612	612	612	612	612	612
$\bar{R}^2(\%)$	1	0	0	0	0	0	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the LLB trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{LLB} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Net external financing, Net debt financing, change in ppe and inv/assets, Change in financial liabilities, Inventory Growth, Asset growth. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 197306 to 202406.

Intercept	0.16 [2.35]	0.17 [2.45]	0.18 [2.60]	0.17 [2.43]	0.18 [2.64]	0.18 [2.56]	0.16 [2.33]
Anomaly 1	13.63 [3.97]						11.22 [2.96]
Anomaly 2		12.12 [3.22]					5.95 [1.15]
Anomaly 3			7.94 [2.55]				3.17 [0.95]
Anomaly 4				9.04 [2.31]			3.34 [0.64]
Anomaly 5					6.09 [2.26]		4.63 [1.63]
Anomaly 6						4.02 [1.00]	-2.12 [-0.51]
mkt	-0.56 [-0.34]	-2.41 [-1.52]	-2.54 [-1.60]	-2.23 [-1.40]	-2.54 [-1.60]	-2.44 [-1.53]	-0.91 [-0.55]
smb	5.01 [1.91]	-0.44 [-0.18]	0.66 [0.27]	-0.29 [-0.12]	1.30 [0.54]	0.34 [0.14]	4.03 [1.44]
hml	-7.64 [-2.50]	-8.66 [-2.83]	-10.01 [-3.25]	-8.55 [-2.78]	-9.35 [-3.05]	-9.33 [-3.03]	-7.87 [-2.53]
rmw	-1.76 [-0.47]	5.06 [1.62]	6.54 [2.10]	5.58 [1.78]	7.27 [2.32]	6.41 [2.05]	-0.55 [-0.14]
cma	15.66 [3.05]	21.39 [4.51]	18.26 [3.47]	21.50 [4.45]	19.12 [3.66]	19.66 [2.92]	10.54 [1.51]
umd	1.39 [0.89]	0.70 [0.44]	1.46 [0.93]	0.70 [0.44]	0.86 [0.54]	1.57 [0.99]	0.19 [0.12]
# months	612	612	612	612	612	612	612
$\bar{R}^2(\%)$	9	8	8	8	8	7	10

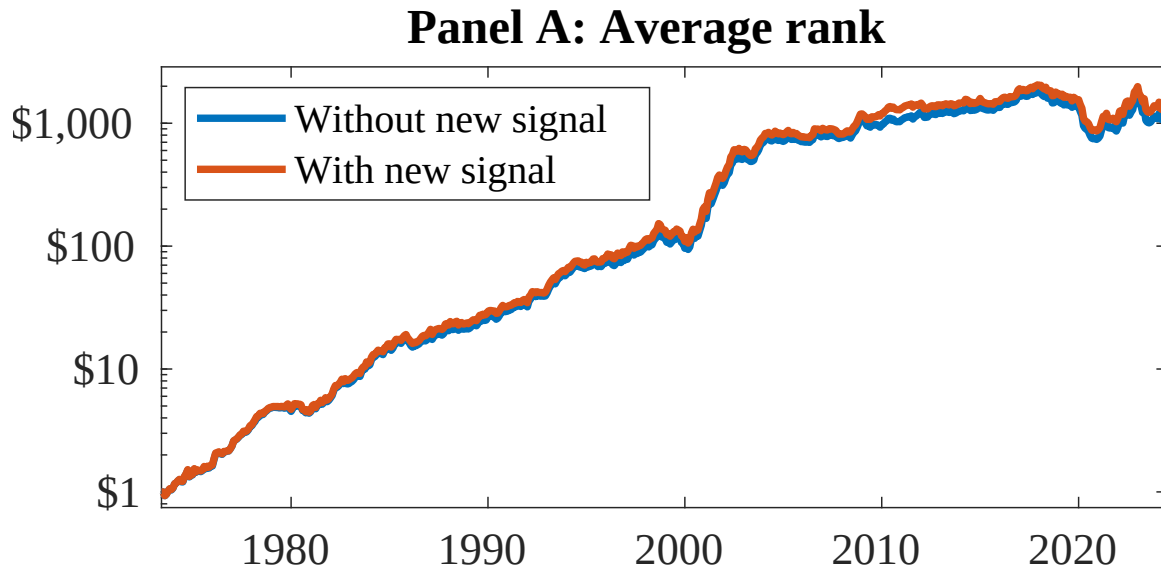


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 155 anomalies. The red solid lines indicate combination trading strategies that utilize the 155 anomalies as well as LLB. Panel A shows results using "Average rank" as the combination method. See Section 7 for details on the combination methods.

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